



Department of Computer Engineering

Academic Year: 2025-26

Year: SE

Semester: IV

Subject: DBMS Lab

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Date of Performance: 10-04-2026

Date of Submission: 13-04-2026

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Experiment No. 10

Aim: Implementation and demonstration of Transaction and Concurrency control techniques using locks.

Output:

```
mysql> LOCK TABLE employee READ;  
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SELECT * FROM employee;
```

fname	lname	salary
Rahul	Sharma	60000
Amit	Patel	65000
Priya	Iyer	60000
Sneha	Kulkarni	48000
Arjun	Reddy	120000
smita	patil	98000

```
6 rows in set (0.00 sec)
```

```
mysql>
```

```
mysql> use exp9;  
Database changed
```

```
mysql> SELECT * FROM employee;
```

fname	lname	salary
Rahul	Sharma	60000
Amit	Patel	65000
Priya	Iyer	60000
Sneha	Kulkarni	48000
Arjun	Reddy	120000
smita	patil	98000

```
6 rows in set (0.00 sec)
```

```
mysql>
```



```
| Arjun | Reddy | 120000 |  
| smita | patil | 98000 |  
+-----+-----+-----+  
6 rows in set (0.00 sec)  
  
mysql> UNLOCK TABLES;  
Query OK, 0 rows affected (0.00 sec)  
  
mysql> LOCK TABLE employee WRITE;  
Query OK, 0 rows affected (0.00 sec)  
  
mysql>
```

```
| Rahul | Sharma | 60000 |  
| Amit | Patel | 65000 |  
| Priya | Iyer | 60000 |  
| Sneha | Kulkarni | 48000 |  
| Arjun | Reddy | 120000 |  
| smita | patil | 98000 |  
+-----+-----+-----+  
6 rows in set (0.00 sec)  
  
mysql> SELECT * FROM employee;  
|
```

```
mysql> UPDATE employee SET salary=95000 WHERE lname='Sharma';  
Query OK, 1 row affected (0.01 sec)  
Rows matched: 1 Changed: 1 Warnings: 0  
  
mysql> unlock tables;  
Query OK, 0 rows affected (0.00 sec)  
  
mysql>
```

```
| Priya | Iyer | 60000 |  
| Sneha | Kulkarni | 48000 |  
| Arjun | Reddy | 120000 |  
| smita | patil | 98000 |  
+-----+-----+-----+  
6 rows in set (2 min 6.38 sec)  
  
mysql> |
```

Conclusion: This experiment enhanced understanding of transaction and concurrency control using locks, ensuring data consistency and preventing conflicts in the database system.